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Just as in the rules for the bathtub, whenever the furnace is putting in more heat than is leaking out, the room temperature rises. Whenever the inflow rate falls behind the outflow rate, the temperature falls. If you study the system changes on this graph for a while and relate them to the feedback-loop diagram of this system, you'll get a good sense of how the structural interconnections of this system—its two feedback loops and how they shift in strength relative to each other—lead to the unfolding of the system's behavior over time.

A Stock with One Reinforcing Loop and One Balancing Loop—Population and Industrial Economy

What happens when a reinforcing and a balancing loop are both pulling on the same stock? This is one of the most common and important system structures. Among other things, it describes every living population and every economy.

A population has a reinforcing loop causing it to grow through its birth rate, and a balancing loop causing it to die off through its death rate.

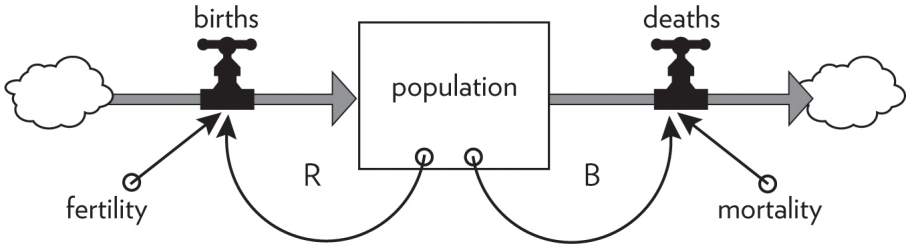


Figure 21. Population governed by a reinforcing loop of births and a balancing loop of deaths.

As long as fertility and mortality are constant (which in real systems they rarely are), this system has a simple behavior. It grows exponentially or dies off, depending on whether its reinforcing feedback loop determining births is stronger than its balancing feedback loop determining deaths.

For example, the 2007 world population of 6.6 billion people had a fertility rate of roughly 21 births a year for every 1,000 people in the population. Its mortality rate was 9 deaths a year out of every 1,000 people. Fertility was higher than mortality, so the reinforcing loop dominated the system. If those fertility and mortality rates continue unchanged, a child born now will see the world population more than double by the time he or she reaches the age of 60, as shown in Figure 22.

If, because of a terrible disease, the mortality rate were higher, say at 30 deaths per 1,000, while the fertility rate remained at 21, then the death loop would dominate the system. More people would die each year than would be born, and the population would gradually decrease (Figure 23).

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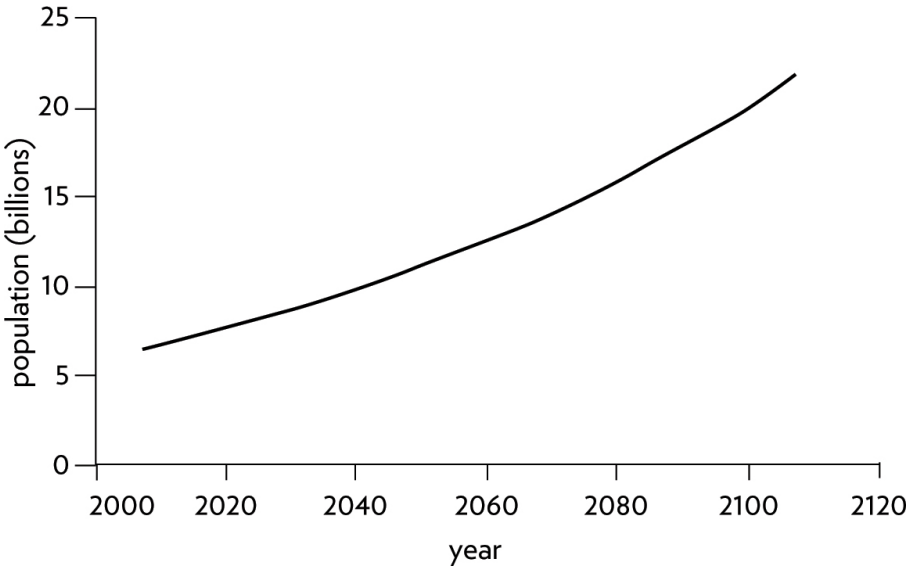


Figure 22. Population growth if fertility and mortality maintain their 2007 levels of 21 births and nine deaths per 1,000 people.

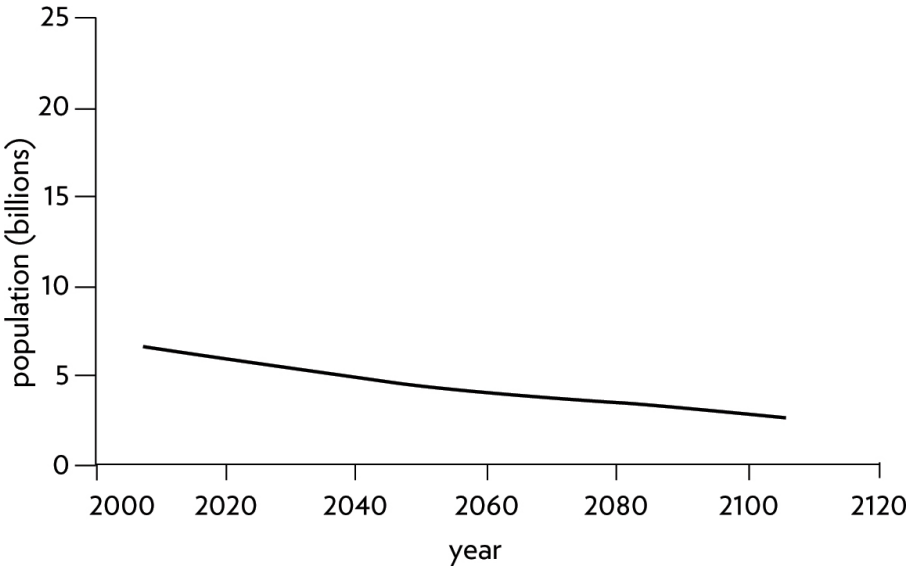


Figure 23. Population decline if fertility remains at 2007 level (21 births per 1,000) but mortality is much higher, 30 deaths per 1,000.

Things get more interesting when fertility and mortality change over time. When the United Nations makes long-range population projections, it generally assumes that, as countries become more developed, average fertility will decline (approaching replacement where on average each woman has 1.85 children). Until recently, assumptions about mortality were that it would also decline, but more slowly (because it is already low in most parts of the world). However, because of the epidemic of HIV/AIDS, the UN now assumes the trend of increasing life expectancy over the next fifty years will slow in regions affected by HIV /AIDS.

Changing flows (fertility and mortality) create a change in the behavior over time of the stock (population)—the line bends. If, for example, world fertility falls steadily to equal mortality by the year 2035 and they both stay constant thereafter, the population will level off, births exactly balancing deaths in dynamic equilibrium, as in Figure 24.

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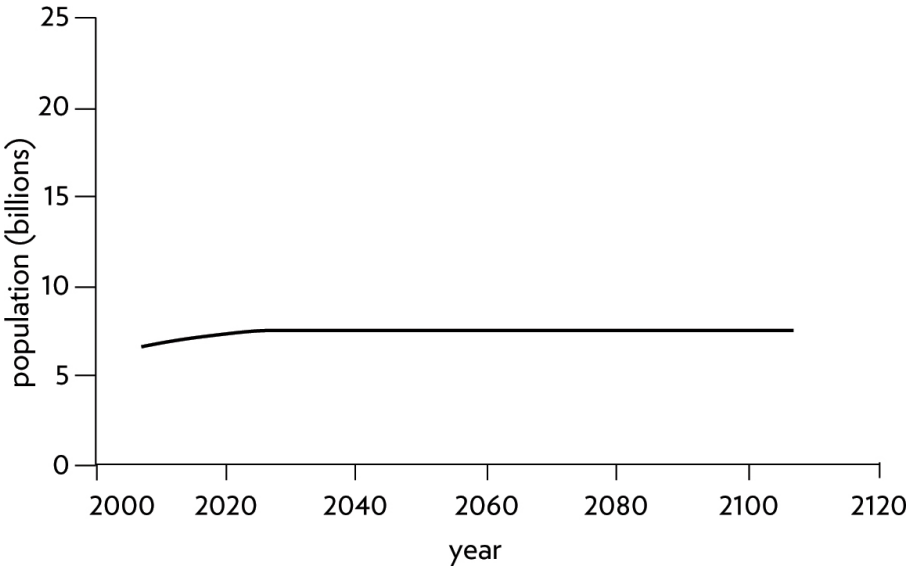


Figure 24. Population stabilizes when fertility equals mortality.

This behavior is an example of **shifting dominance** of feedback loops. Dominance is an important concept in systems thinking. When one loop dominates another, it has a stronger impact on behavior. Because systems often have several competing feedback loops operating simultaneously, those loops that dominate the system will determine the behavior.

At first, when fertility is higher than mortality, the reinforcing growth loop dominates the system and the resulting behavior is exponential growth. But that loop is gradually weakened as fertility falls. Finally, it exactly equals the strength of the balancing loop of mortality. At that point neither loop dominates, and we have dynamic equilibrium.

You saw shifting dominance in the thermostat system, when the outside temperature fell and the heat leaking out of the poorly insulated house overwhelmed the ability of the furnace to put heat into the room. Dominance shifted from the heating loop to the cooling loop.

Complex behaviors of systems often arise as the relative strengths of feedback loops shift, causing first one loop and then another to dominate behavior.

There are only a few ways a population system could behave, and these depend on what happens to the "driving" variables, fertility and mortality. These are the only ones possible for a simple system of one reinforcing and one balancing loop. A stock governed by linked reinforcing and balancing loops will grow exponentially if the reinforcing loop dominates the balancing one. It will die off if the balancing loop dominates the reinforcing one. It will level off if the two loops are of equal strength (see Figure 25). Or it will do a sequence of these things, one after another, if the relative strengths of the two loops change over time (see Figure 26).

I chose some provocative population scenarios here to illustrate a point about models and the scenarios they can generate. Whenever you are confronted with a scenario (and you are, every time you hear about an economic prediction, a corporate budget, a weather forecast, future climate change, a stockbroker saying what is going to happen to a particular holding), there are questions you need to ask that will help you decide how good a representation of reality is the underlying model.

- Are the driving factors likely to unfold this way? (What are birth rate and death rate likely to do?)
- If they did, would the system react this way? (Do birth and death rates really cause the population stock to behave as we think it will?)

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- What is driving the driving factors? (What affects birth rate? What affects death rate?)

The first question can't be answered factually. It's a guess about the future, and the future is inherently uncertain. Although you may have a strong opinion about it, there's no way to prove you're right until the future actually happens. A systems analysis can test a number of scenarios to see what happens if the driving factors do different things. That's usually one purpose of a systems analysis. But you have to be the judge of which scenario, if any, should be taken seriously as a future that might really be possible.

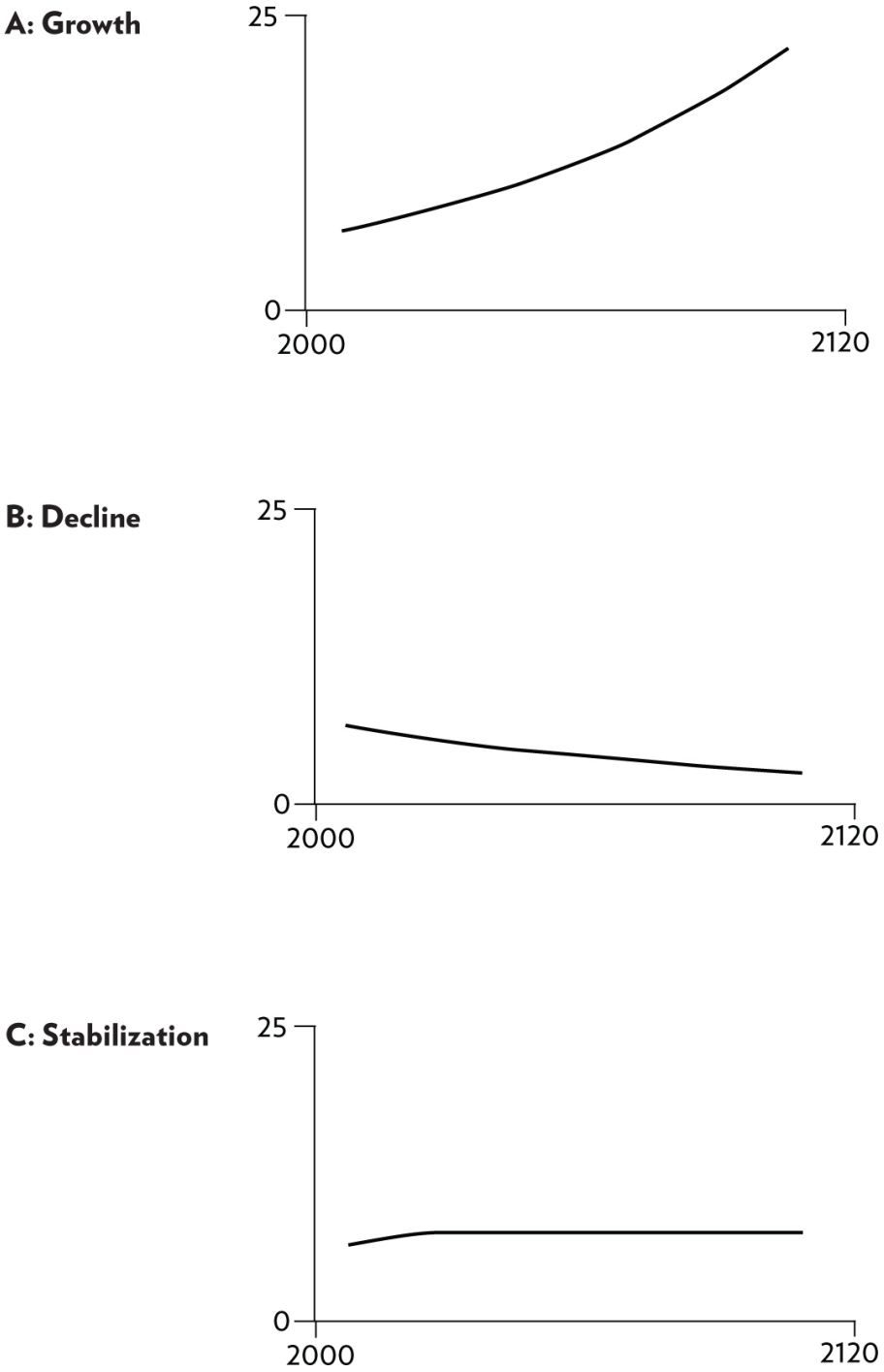


Figure 25. Three possible behaviors of a population: growth, decline, and stabilization.

Dynamic systems studies usually are not designed to *predict* what will happen. Rather, they're designed to explore *what would happen*, if a number of driving factors unfold in a range of different ways.

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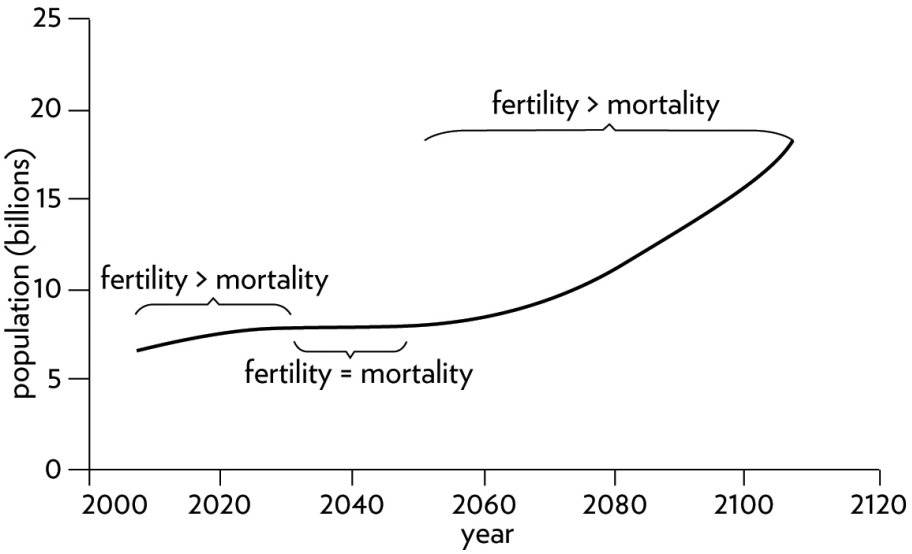


Figure 26. Shifting dominance of fertility and mortality loops.

The second question—whether the system really will react this way—is more scientific. It's a question about how good the model is. Does it capture the inherent dynamics of the system? Regardless of whether you think the driving factors *will* do that, *would the system behave like that* if they did?

In the population scenarios above, however likely you think they are, the answer to this second question is roughly yes, the population would behave like this, if the fertility and mortality did that. The population model I have used here is very simple. A more detailed model would distinguish age groups, for example. But basically this model responds the way a real population would, growing under the conditions when a real population would grow, declining when a real population would decline. The numbers are off, but the basic behavior pattern is realistic.

System dynamics models explore possible futures and ask "what if" questions.

Finally, there is the third question. What is driving the driving factors? What is adjusting the inflows and outflows? This is a question about system boundaries. It requires a hard look at those driving factors to see if they are actually independent, or if they are also embedded in the system.

QUESTIONS FOR TESTING THE VALUE OF A MODEL

1. Are the driving factors likely to unfold this way?
2. If they did, would the system react this way?
3. What is driving the driving factors?

Is there anything about the size of the population, for instance, that might feed back to influence fertility or mortality? Do other factors—economics, the environment, social trends—influence fertility and mortality? Does the size of the population affect those economic and environmental and social factors?

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Model utility depends not on whether its driving scenarios are realistic (since no one can know that for sure), but on whether it responds with a realistic pattern of behavior.

Of course, the answer to all of these questions is yes. Fertility and mortality are governed by feedback loops too. At least some of those feedback loops are themselves affected by the size of the population. This population "animal" is only one piece of a much larger system.[3]

One important piece of the larger system that affects population is the economy. At the heart of the economy is another reinforcing-loop-plus-balancing-loop system—the same kind of structure, with the same kinds of behavior, as the population (see Figure 27).

The greater the stock of physical capital (machines and factories) in the economy and the efficiency of production (output per unit of capital), the more output (goods and services) can be produced each year.

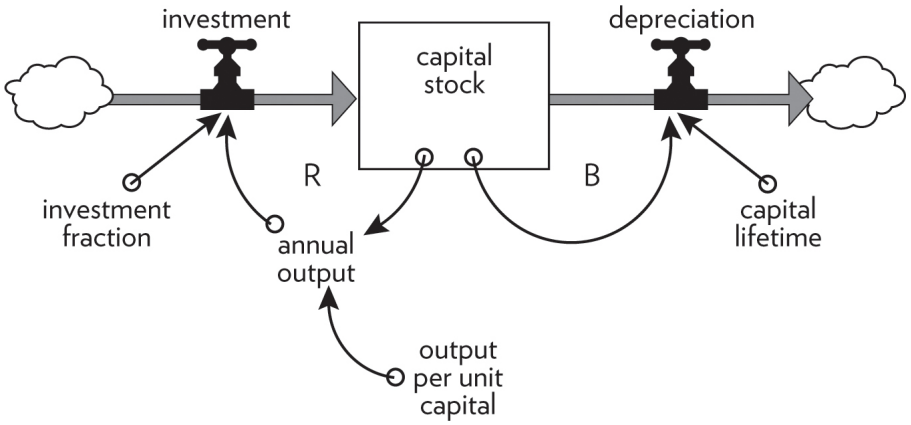


Figure 27. Like a living population, economic capital has a reinforcing loop (investment of output) governing growth and a balancing loop (depreciation) governing decline.

The more output that is produced, the more can be invested to make new capital. This is a reinforcing loop, like the birth loop for a population. The investment fraction is equivalent to the fertility. The greater the fraction of its output a society invests, the faster its capital stock will grow.

Physical capital is drained by depreciation—obsolescence and wearing-out. The balancing loop controlling depreciation is equivalent to the death loop in a population. The "mortality" of capital is determined by the average capital lifetime. The longer the lifetime, the smaller the fraction of capital that must be retired and replaced each year.

If this system has the same structure as the population system, it must have the same repertoire of behaviors. Over recent history world capital, like world population, has been dominated by its reinforcing loop and has been growing exponentially. Whether in the future it grows or stays constant or dies off depends on whether its reinforcing growth loop remains stronger than its balancing depreciation loop. That depends on

- the investment fraction—how much output the society invests rather than consumes,
- the efficiency of capital—how much capital it takes to produce a given amount of output, and
- the average capital lifetime.

If a constant fraction of output is reinvested in the capital stock and the efficiency of that capital (its ability to produce output) is also constant, the capital stock may decline, stay constant, or grow, depending on the lifetime of the capital. The lines in Figure 28 show systems with different average capital lifetimes. With a relatively short lifetime, the capital wears out faster than it is replaced. Reinvestment does not keep up with depreciation and the economy slowly declines. When depreciation just balances investment, the

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economy is in dynamic equilibrium. With a long lifetime, the capital stock grows exponentially. The longer the lifetime of capital, the faster it grows.

This is another example of a principle we've already encountered: You can make a stock grow by decreasing its outflow rate as well as by increasing its inflow rate!

Just as many factors influence the fertility and mortality of a population, so many factors influence the output ratio, investment fraction, and the lifetime of capital—interest rates, technology, tax policy, consumption habits, and prices, to name just a few. Population itself influences investment, both by contributing labor to output, and by increasing demands on consumption, thereby decreasing the investment fraction. Economic output also feeds back to influence population in many ways. A richer economy usually has better health care and a lower death rate. A richer economy also usually has a lower birth rate.

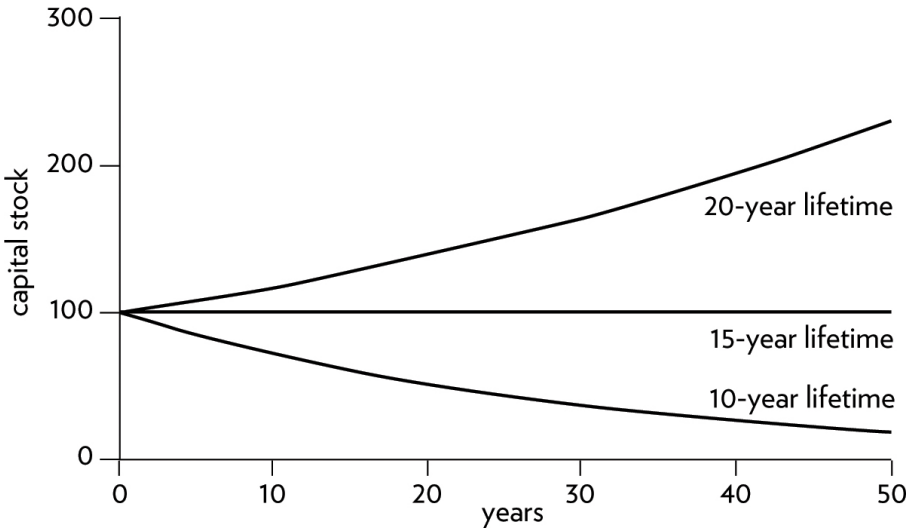


Figure 28. Growth in capital stock with changes in the lifetime of the capital. In a system with output per unit capital ratio of 1:3 and an investment fraction of 20 percent, capital with a lifetime of 15 years just keeps up with depreciation. A shorter lifetime leads to a declining stock of capital.

In fact, just about any long-term model of a real economy should link together the two structures of population and capital to show how they affect each other. The central question of economic development is how to keep the reinforcing loop of capital accumulation from growing more slowly than the reinforcing loop of population growth—so that people are getting richer instead of poorer.[4]

It may seem strange to you that I call the capital system the same kind of "zoo animal" as the population system. A production system with factories and shipments and economic flows doesn't look much like a population system with babies being born and people aging and having more babies and dying. But from a systems point of view these systems, so dissimilar in many ways, have one important thing in common: their feedback-loop structures. Both have a stock governed by a reinforcing growth loop and a balancing death loop. Both also have an aging process. Steel mills and lathes and turbines get older and die just as people do.

Systems with similar feedback structures produce similar dynamic behaviors.

One of the central insights of systems theory, as central as the observation that systems largely cause their own behavior, is that systems with similar feedback structures produce similar dynamic behaviors, even if the outward appearance of these systems is completely dissimilar.

A population is nothing like an industrial economy, except that both can reproduce themselves out of themselves and thus grow exponentially. And both age and die. A coffee cup cooling is like a warmed room cooling, and like a radioactive substance decaying, and like a population or industrial economy aging and dying. Each declines as the result of a balancing feedback loop.

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A System with Delays—Business Inventory

Picture a stock of inventory in a store—a car dealership—with an inflow of deliveries from factories and an outflow of new car sales. By itself, this stock of cars on the dealership lot would behave like the water in a bathtub.

Now picture a regulatory feedback system designed to keep the inventory high enough so that it can always cover ten days' worth of sales (see Figure 29). The car dealer needs to keep some inventory because deliveries and purchases don't match perfectly every day. Customers make purchases that are unpredictable on a day-to-day basis. The car dealer also needs to provide herself with some extra inventory (a buffer) in case deliveries from suppliers are delayed occasionally.

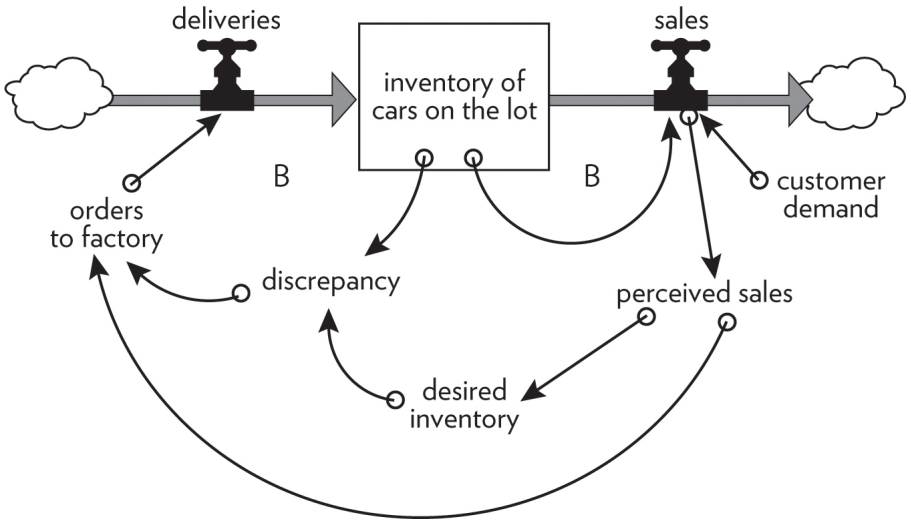


Figure 29. Inventory at a car dealership is kept steady by two competing balancing loops, one through sales and one through deliveries.

The dealer monitors sales (perceived sales), and if, for example, they seem to be rising, she adjusts orders to the factory to bring inventory up to her new desired inventory that provides ten days' coverage at the higher sales rate. So, higher sales mean higher perceived sales, which means a higher discrepancy between inventory and desired inventory, which means higher orders, which will bring in more deliveries, which will raise inventory so it can comfortably supply the higher rate of sales.

This system is a version of the thermostat system—one balancing loop of sales draining the inventory stock and a competing balancing loop maintaining the inventory by resupplying what is lost in sales. Figure 30 shows the not very surprising result of an increase in consumer demand of 10 percent.

In Figure 31, I am putting something else into this simple model—three delays that are typical of what we experience in the real world.

First, there is a perception delay, intentional in this case. The car dealer doesn't react to just any blip in sales. Before she makes ordering decisions, she averages sales over the past five days to sort out real trends from temporary dips and spikes.

Second, there is a response delay. Even when it's clear that orders need to be adjusted, she doesn't try to make up the whole adjustment in a single order. Rather, she makes up one-third of any shortfall with each order. Another way of saying that is, she makes partial adjustments over three days to be extra sure the trend is real. Third, there is a delivery delay. It takes five days for the supplier at the factory to receive an order, process it, and deliver it to the dealership.

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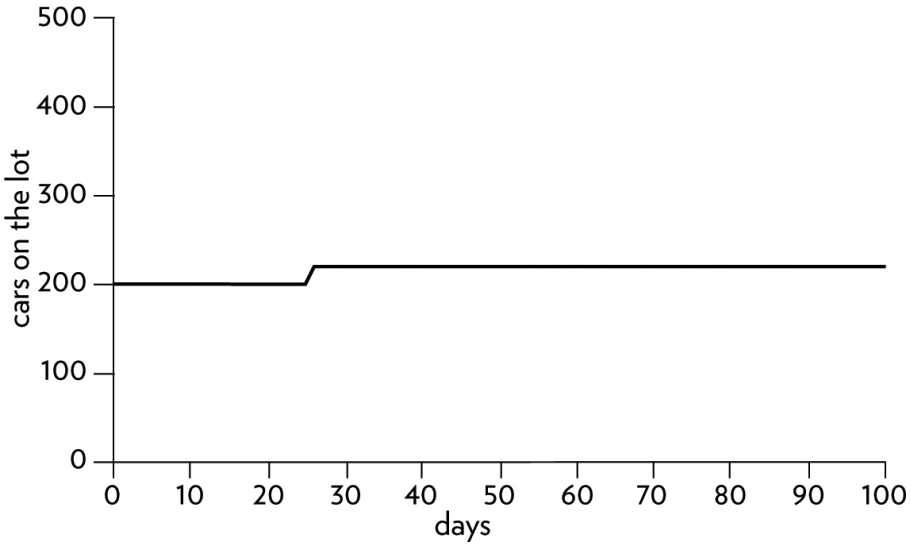


Figure 30. Inventory on the car dealership's lot with a permanent 10-percent increase in consumer demand starting on day 25.

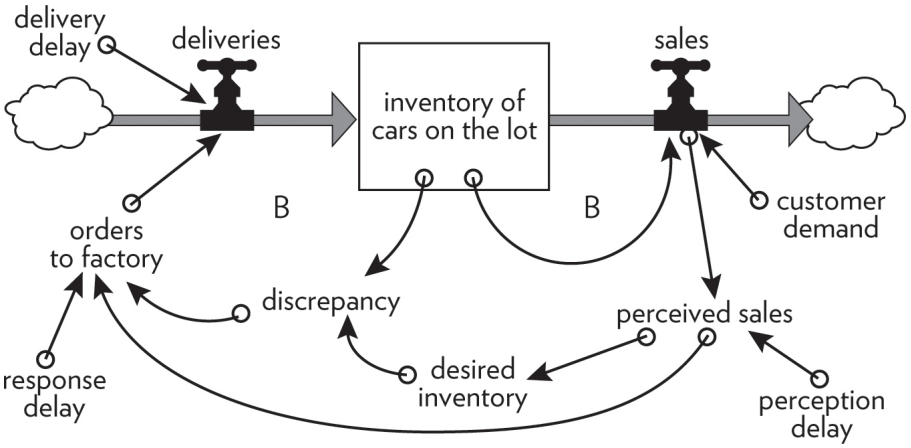


Figure 31. Inventory at a car dealership with three common delays now included in the picture—a perception delay, a response delay, and a delivery delay.

Although this system still consists of just two balancing loops, like the simplified thermostat system, it doesn't behave like the thermostat system. Look at what happens, for example, as shown in Figure 32, when the business experiences the same permanent 10-percent jump in sales from an increase in customer demand.

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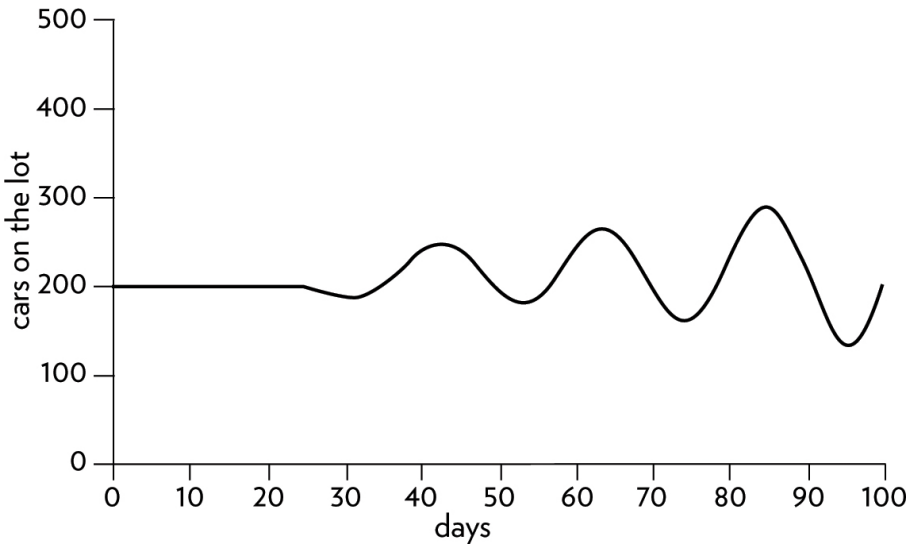


Figure 32. Response of inventory to a 10-percent increase in sales when there are delays in the system.

Oscillations! A single step up in sales causes inventory to drop. The car dealer watches long enough to be sure the higher sales rate is going to last. Then she begins to order more cars to both cover the new rate of sales and bring the inventory up. But it takes time for the orders to come in. During that time inventory drops further, so orders have to go up a little more, to bring inventory back up to ten days' coverage.

Eventually, the larger volume of orders starts arriving, and inventory recovers—and more than recovers, because during the time of uncertainty about the actual trend, the owner has ordered too much. She now sees her mistake, and cuts back, but there are still high past orders coming in, so she orders even less. In fact, almost inevitably, since she still can't be sure of what is going to happen next, she orders too little. Inventory gets too low again. And so forth, through a series of oscillations around the new desired inventory level. As Figure 33 illustrates, what a difference a few delays make!

We'll see in a moment that there are ways to damp these oscillations in inventory, but first it's important to understand why they occur. It isn't because the car dealer is stupid. It's because she is struggling to operate in a system in which she doesn't have, and can't have, timely information and in which physical delays prevent her actions from having an immediate effect on inventory. She doesn't know what her customers will do next. When they do something, she's not sure they'll keep doing it. When she issues an order, she doesn't see an immediate response. This situation of information insufficiency and physical delays is very common. Oscillations like these are frequently encountered in inventories and in many other systems. Try taking a shower sometime where there's a very long pipe between the hot- and cold-water mixer and the showerhead, and you'll experience directly the joys of hot and cold oscillations because of a long response delay.

A delay in a balancing feedback loop makes a system likely to oscillate.

How much of a delay causes what kind of oscillation under what circumstances is not a simple matter. I can use this inventory system to show you why.

"These oscillations are intolerable," says the car dealer (who is herself a learning system, determined now to change the behavior of the inventory system). "I'm going to shorten the delays. There's not much I can do about the delivery delay from the factory, so I'm going to react faster myself. I'll average sales trends over only two days instead of five before I make order adjustments."

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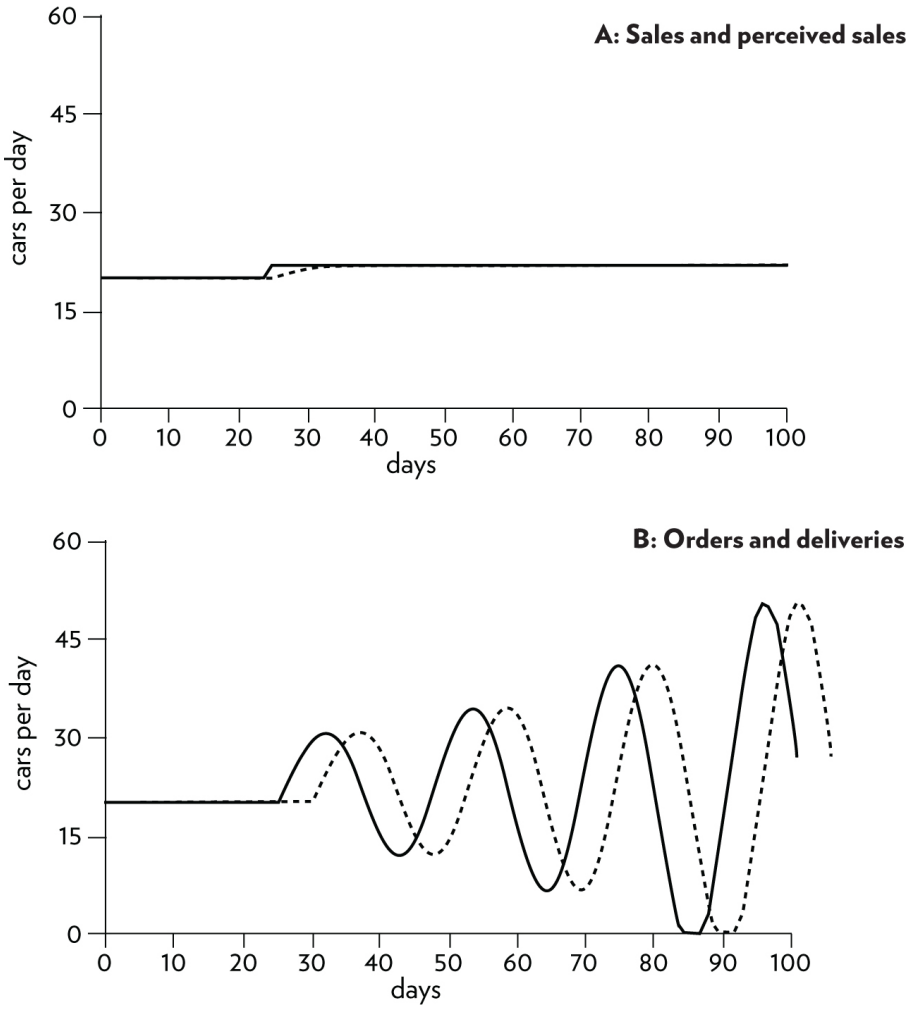


Figure 33. The response of orders and deliveries to an increase in demand. *A* shows the small but sharp step up in sales on day 25 and the car dealer's "perceived" sales, in which she averages the change over 3 days. *B* shows the resulting ordering pattern, tracked by the actual deliveries from the factory.

Figure 34 illustrates what happens when the dealer's perception delay is shortened from five days to two.

Not much happens when the car dealer shortens her perception delay. If anything the oscillations in the inventory of cars on the lot are a bit worse. And if, instead of shortening her perception time, the car dealer tries shortening her reaction time—making up perceived shortfalls in two days instead of three—things get very much worse, as shown in Figure 35.

Something has to change and, since this system has a learning person within it, something will change. "High leverage, wrong direction," the system-thinking car dealer says to herself as she watches this failure of a policy intended to stabilize the oscillations. This perverse kind of result can be seen all the time—someone trying to fix a system is attracted intuitively to a policy lever that in fact does have a strong effect on the system. And then the well-intentioned fixer pulls the lever in the wrong direction! This is just one example of how we can be surprised by the counterintuitive behavior of systems when we start trying to change them.

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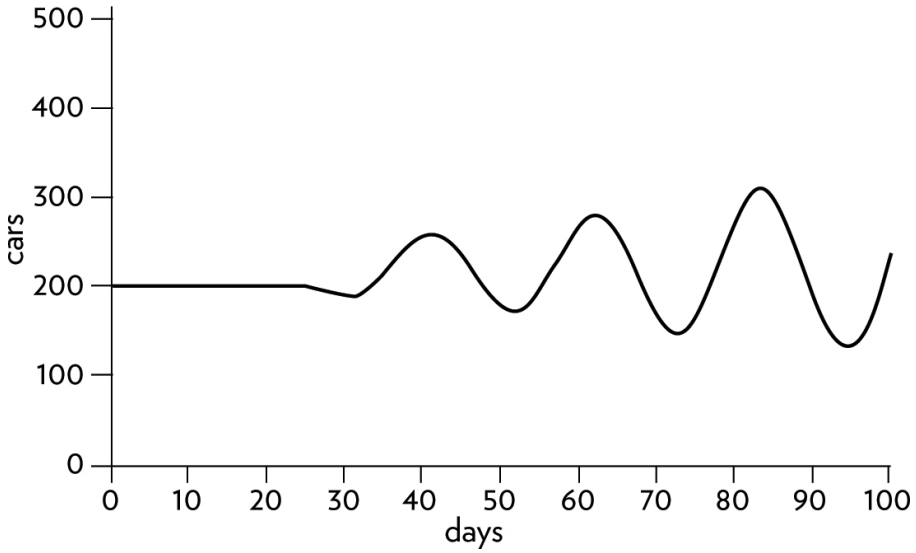


Figure 34. The response of inventory to the same increase in demand with a shortened perception delay.

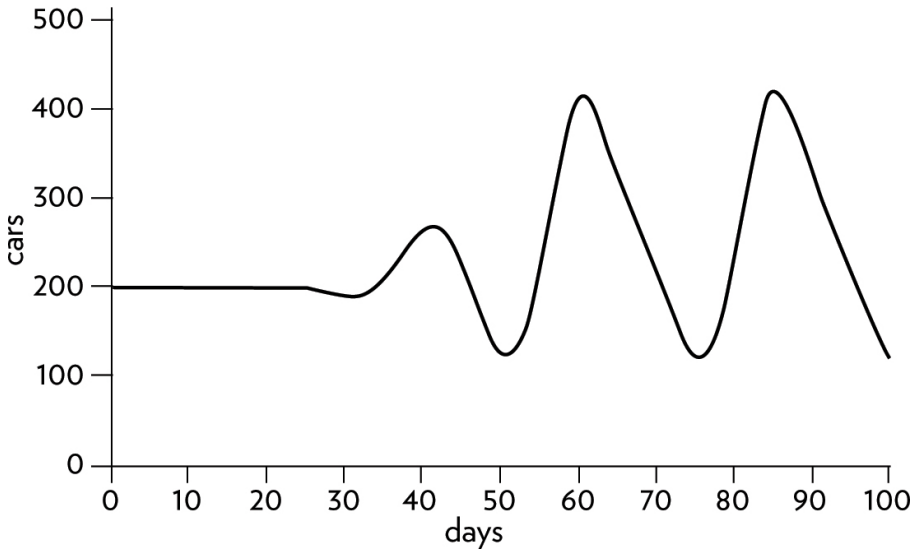


Figure 35. The response of inventory to the same increase in demand with a shortened reaction time. Acting faster makes the oscillations worse!

Part of the problem here is that the car dealer has been reacting not too slowly, but too quickly. Given the configuration of this system, she has been overreacting. Things would go better if, instead of decreasing her response delay from three days to two, she would increase the delay from three days to six, as illustrated in Figure 36.

As Figure 36 shows, the oscillations are greatly damped with this change, and the system finds its new equilibrium fairly efficiently.

The most important delay in this system is the one that is not under the direct control of the car dealer. It's the delay in delivery from the factory. But even without the ability to change that part of her system, the dealer can learn to manage inventory quite well.

Delays are pervasive in systems, and they are strong determinants of behavior. **Changing the length of a delay may** (or may not, depending on the type of delay and the relative lengths of other delays) **make a large change in the behavior of a system.**

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Changing the delays in a system can make it much easier or much harder to manage. You can see why system thinkers are somewhat fanatic on the subject of delays. We're always on the alert to see where delays occur in systems, how long they are, whether they are delays in information streams or in physical processes. We can't begin to understand the dynamic behavior of systems unless we know where and how long the delays are. And we are aware that some delays can be powerful policy levers. Lengthening or shortening them can produce major changes in the behavior of systems.

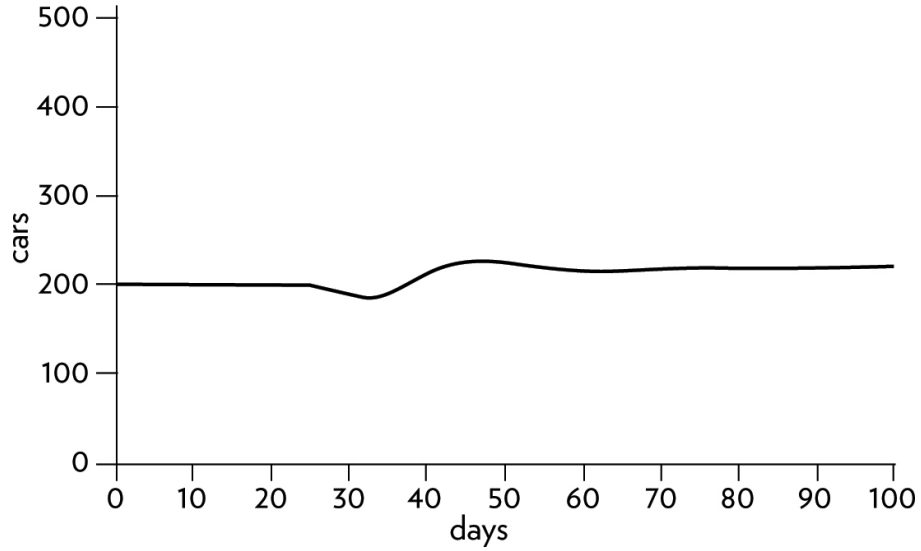


Figure 36. The response of inventory to the same increase in demand with a slowed reaction time.

In the big picture, one store's inventory problem may seem trivial and fixable. But imagine that the inventory is that of all the unsold automobiles in America. Orders for more or fewer cars affect production not only at assembly plants and parts factories, but also at steel mills, rubber and glass plants, textile producers, and energy producers. Everywhere in this system are perception delays, production delays, delivery delays, and construction delays. Now consider the link between car production and jobs—increased production increases the number of jobs allowing more people to buy cars. That's a reinforcing loop, which also works in the opposite direction—less production, fewer jobs, fewer car sales, less production. Put in another reinforcing loop, as speculators buy and sell shares in the auto and auto-supply companies based on their recent performance, so that an upsurge in production produces an upsurge in stock price, and vice versa.

That very large system, with interconnected industries responding to each other through delays, entraining each other in their oscillations, and being amplified by multipliers and speculators, is the primary cause of business cycles. Those cycles don't come from presidents, although presidents can do much to ease or intensify the optimism of the upturns and the pain of the downturns. Economies are extremely complex systems; they are full of balancing feedback loops with delays, and they are inherently oscillatory.[5]

Two-Stock Systems

A Renewable Stock Constrained by a Nonrenewable Stock—an Oil Economy

The systems I've displayed so far have been free of constraints imposed by their surroundings. The capital stock of the industrial economy model didn't require raw materials to produce output. The population didn't need food. The thermostat-furnace system never ran out of oil. These simple models of the systems have been able to operate according to their unconstrained internal dynamics, so we could see what those dynamics are.

But any real physical entity is always surrounded by and exchanging things with its environment. A corporation needs a constant supply of energy and materials and workers and managers and customers. A growing corn crop needs water and nutrients and protection from pests. A population needs food and water

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